CS5330 · FINAL PROJECT · SPRING 2026

Real-Time Object Recognition and AR Overlay

Using deep learning to turn any everyday object into an augmented reality anchor

Sangeeth Deleep Menon · NUID 002524579
Khoury College of Computer Sciences · Northeastern University

THE PROBLEM

Traditional AR needs a printed marker. What if the object itself was the marker?

PROJECT 4 GAVE US

AR rendering

3D graphics overlaid using a chessboard calibration target

PROJECT 5 GAVE US

Deep learning

CNNs and Vision Transformers trained to classify images

This project combines both — no marker required

SYSTEM OVERVIEW

A three-step pipeline

1. **Classify** — crop a center ROI from the webcam frame and run it through MobileNetV2 to get a class and confidence score
2. **Estimate pose** — use a fixed-depth pinhole camera model to back-project the ROI bounding box into 3D space at 0.5 m
3. **Render** — project 8 box corners to pixels with `cv2.projectPoints` and draw a 3D wireframe box with a floating label tag

DATASET

10 classes · 1419 images

CLASS	IMAGES	SOURCE
book	161	Webcam
cup	173	Webcam
keyboard	152	Webcam
pen	171	Webcam
phone	159	Webcam
glasses	130	Web
headphones	110	Web
laptop	123	Web
ps5_controller	102	Web
tablet	138	Web

TRAIN / VAL / TEST SPLIT

70% 15% 15%

Fixed seed 42 — fully reproducible

AUGMENTATION (TRAIN ONLY)

Horizontal flip · Color jitter · ±15° rotation

KEY OBSERVATION

Webcam classes outperform web-only classes — data quality matters more than quantity

MODEL ARCHITECTURES

Three approaches, one winner

CUSTOM CNN

ObjectCNN

- 3-layer conv network
- 64×64 RGB input
- BatchNorm + Dropout 0.4
- Trained from scratch
- ~1.2M parameters

~100% on 5 classes

VISION TRANSFORMER

ObjectViT

- 8×8 patch embedding
- 64 patches per image
- 4 transformer layers
- Trained from scratch
- 128-dim embeddings

Slower to converge

TRANSFER LEARNING ✦ DEPLOYED

MobileNetV2

- ImageNet pretrained
- 224×224 RGB input
- Fine-tuned classifier head
- Dropout 0.3
- Best on noisy web data

84.1% on 10 classes

RESULTS

Numbers that matter

TEST ACCURACY

84.1%

MobileNetV2 on 10-class held-out test set

LIVE FRAME RATE

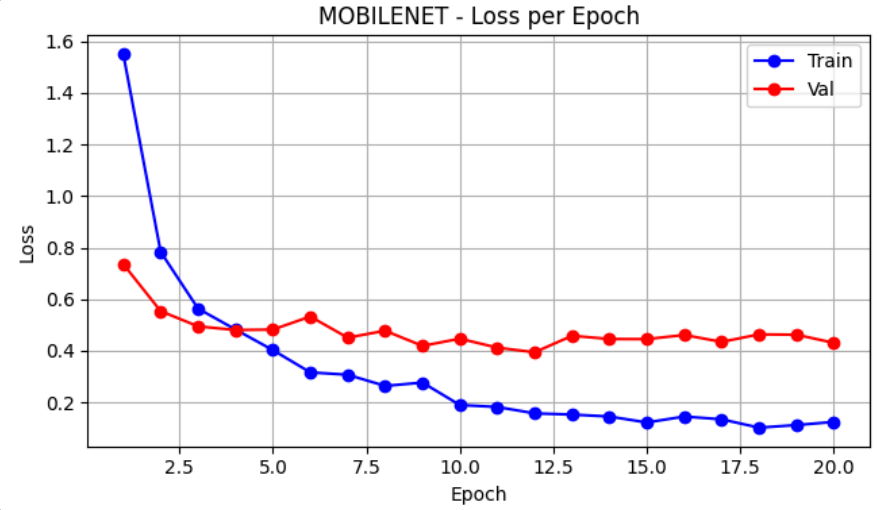
30fps

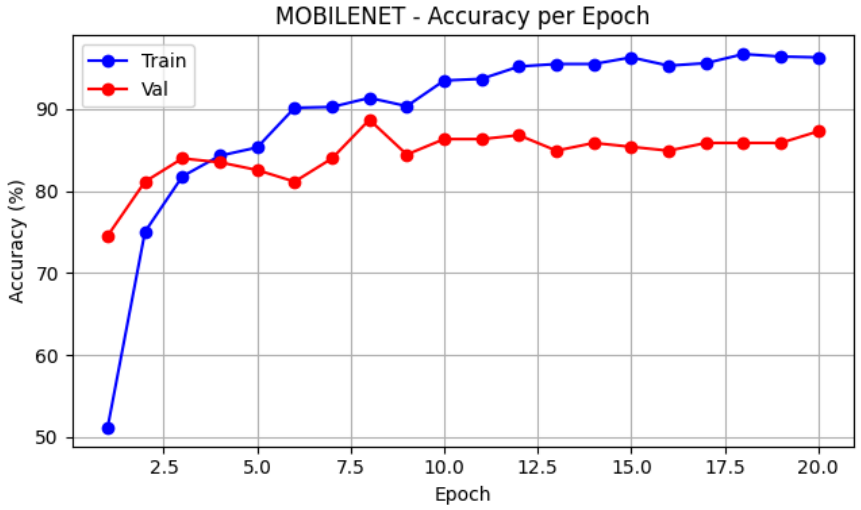
Apple M-series with MPS acceleration

STRONGEST CLASS

Phone

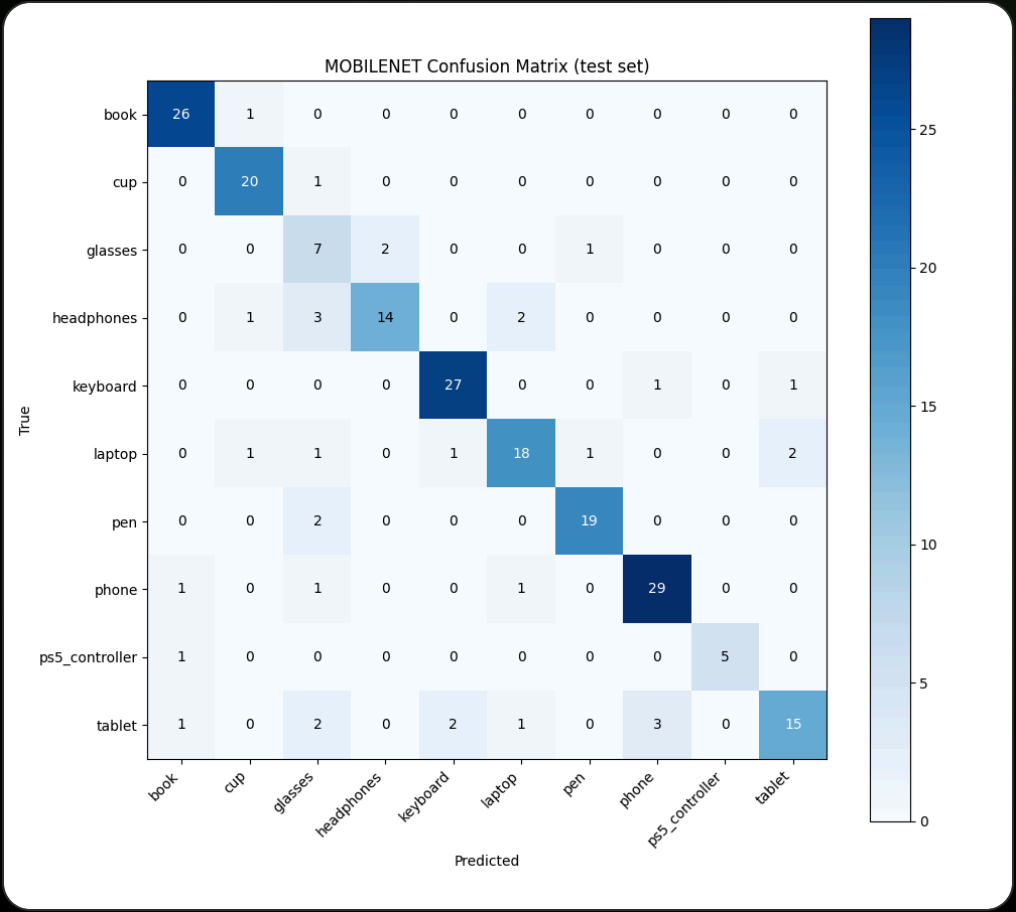
29 of 30 test images correct





WHERE IT STRUGGLES

Confusion tells the story



GLASSES — WEAKEST CLASS

Web crawl mixed eyeglasses with drinking glasses. The model had no way to separate them.

TABLET ↔ PHONE CONFUSION

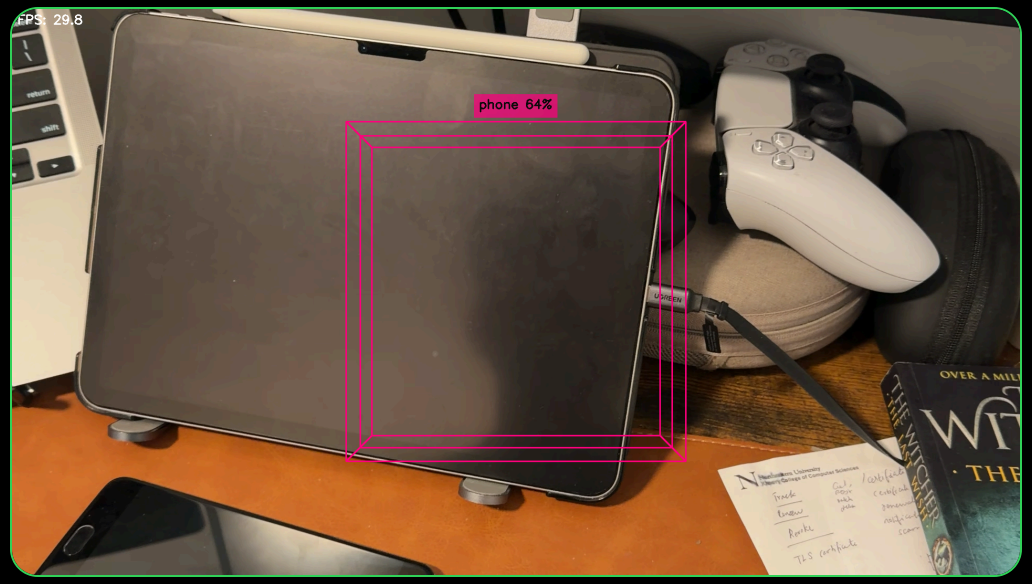
Flat rectangles look alike at certain angles. The demo screenshot shows this live.

ROOT CAUSE

All 5 weak classes used web images only. Webcam-captured classes had near-perfect scores.

LIVE DEMO

AR overlay in action



A tablet is in frame. The model predicts **phone at 84%** confidence.

The 3D wireframe box and label are rendered correctly around the detected region even though the class is wrong — the AR pipeline itself is robust.

● LIVE STATS

29.8 FPS live

0.60 Confidence threshold

LIMITATIONS & NEXT STEPS

Honest about the gaps

CURRENT LIMITATIONS

- **Fixed center ROI** — only detects objects at center of frame
- **Fixed depth (0.5 m)** — box scale is wrong at very close or far distances
- **No background class** — always outputs a class even when nothing is there
- **Prediction flicker** — no temporal smoothing between frames

EASY WINS NEXT

- Collect webcam images for the 5 web-only classes to push accuracy above 90%
- Add a 5-frame history buffer to smooth out prediction flickering
- Add a background class so the system can say "I don't know"
- Replace the fixed ROI with a sliding-window or YOLO-based detector

SUMMARY

Any object the network knows becomes its own AR anchor.

10	84.1%	30fps	0
object classes	test accuracy	live inference	printed markers needed

Sangeeth Deleep Menon · CS5330 Spring 2026 · Northeastern University